AUSM LAB York University · Lassonde School of Engineering



POSTDOCTORAL OPPORTUNITIES

Postdoctoral fellows

Structured Spatial World Models

Toronto, Canada

Sensing · Autonomy

Drones, image–LiDAR fusion, tiny-object detection in orbit

3D World Models

LoD2 buildings, BIM digital twins, Gaussian splatting

Prediction · Action

Mapping agents, LLM and vision-language-action models

Start
Flexible (2026–2027)

Applications
Reviewed on a rolling basis

Email your **CV, research statement and 2–3 papers** to **gsohn@yorku.ca**
Subject: **[Postdoc] Your Name**

ausmlab.com/join

POSTDOCTORAL OPPORTUNITIES

Postdoctoral fellows

Applications reviewed on a rolling basis · Start: Flexible (2026–2027)

Lead research on structured spatial world models, mentor graduate students, and work directly with our industry partners on real-world sensing and mapping problems.

ABOUT THE LAB

The AUSMLab at York University is always looking for talented and motivated researchers to join our team. Our research develops embodied spatial intelligence – AI systems that perceive, reason about, and act within physical space – with applications in smart construction, autonomous infrastructure monitoring, autonomous mapping and navigation, urban digital twins, and AI for Earth and space observation.

We are currently recruiting Ph.D. and M.Sc. students (and welcome inquiries from prospective postdoctoral fellows) supported by active research programs, including the NSERC Discovery Grant “Embodied Spatial Intelligence for Autonomous Infrastructure Monitoring” (2026-2031), the NSERC CREATE “Smart Mobility Advanced Research and Training (SMART)” program, and industry-partnered projects with organizations such as Teledyne Optech, Aethon Aerial Solutions, and the Ontario Ministry of Transportation.

RESEARCH FOCUS

- **Sensing · Autonomy**
  Drones, image–LiDAR fusion, tiny-object detection in orbit
- **3D World Models**
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- **Prediction · Action**
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RESEARCH AREAS

- Neural computer vision and robotic perception
- Autonomous mapping and navigation (SLAM, mobile mapping, UAVs)
- 3D urban modelling and digital twins of the built environment
- Vision-language models for infrastructure and railway monitoring
- AI for Earth and space observation, including Space Situational Awareness

WE OFFER

- Funding through NSERC Discovery, NSERC CREATE SMART and industry partners
- Support to apply for the Canada Postdoctoral Research Award (CPRA) and Mitacs Accelerate fellowships
- Experience leading industry projects and mentoring graduate students
- A GPU computing cluster and real-world data from industry partners

WHO WE LOOK FOR

- A Ph.D. (completed or near completion) in computer vision, machine learning, robotics, photogrammetry and remote sensing, geomatics, or a related field.
- A strong publication record in leading venues (e.g., CVPR, ICCV, ECCV, ISPRS, IEEE TGRS).
- Hands-on experience with deep learning (Python, PyTorch) and 3D or spatial data.

HOW TO APPLY

Email **Prof. Gunho Sohn, gsohn@yorku.ca** with:

- Your CV with a full publication list
- A research statement (up to two pages)
- Two or three representative publications
- Names and contact details of three referees

Subject line: **[Postdoc] Your Name**

FUNDING

Positions are funded through the lab's research grants and industry partners. We also encourage strong candidates to apply with us for external fellowships, such as the Canada Postdoctoral Research Award (CPRA) and Mitacs Accelerate, and we support the application.